

SentinelGrid — One-Page Viva Handout

Case Study 94 · DevOps for National Cyber Defense · Multi-region cloud-native platform

What it does

Ingests **billions of security events/day** from partner orgs across **6 critical-infrastructure sectors**, correlates and scores threats, and distributes actionable intelligence as **STIX/TAXII feeds**. Survives DDoS, AZ failure, ransomware, insider abuse, and full regional outage.

Architecture in 60 seconds

Two AWS regions (active us-east-1, warm us-west-2). Each runs **EKS** with 3 node groups (system, app, ingest). **Six microservices**: `threat-ingestion` → `threat-analysis` → `intel-distribution`, plus `incident-coordination`, `api-gateway` (Kong + OPA), `dashboard-ui`. State: **RDS PostgreSQL Multi-AZ**, **MSK Kafka**, **Elasticsearch**, **S3 with Object Lock**. All wired with **continuous cross-region replication**.

DevOps tooling — case-study requirements mapped 1:1